

Air Quality Index Prediction: A Comparative Study across Diverse Models

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ABSTRACT Accurate forecasting of the Air Quality Index (AQI) has become increasingly critical in India due to rising pollution levels driven by rapid urbanization and industrialization. This study investigates and compares a suite of advanced predictive models—including Long Short-Term Memory (LSTM), Prophet, ARIMA-BiLSTM hybrid, and XGBoost integrated with nature-inspired optimization algorithms such as Particle Swarm Optimization (PSO) and Grey Wolf Optimization (GWO). Using a comprehensive dataset from the Central Pollution Control Board, each model underwent robust preprocessing, feature engineering, and hyperparameter tuning. Among all, the XGBoost-GWO model delivered the highest performance with an R^2 score of 0.9523 and RMSE of 27.09. The results underscore the effectiveness of hybrid modeling and metaheuristic optimization in capturing the nonlinear and seasonal complexities of urban air quality. This work provides a benchmark for state-of-the-art AQI prediction methods and offers a foundation for developing real-time forecasting systems to guide public health interventions and urban policy across Indian cities.

INDEX TERMS Air Quality Index (AQI), ARIMA-BiLSTM, deep learning, Grey Wolf Optimization (GWO), hybrid modeling, LSTM, machine learning, Particle Swarm Optimization (PSO), Prophet, time series forecasting, XGBoost

I. INTRODUCTION

IN recent decades, global air quality has experienced a marked deterioration, with developing countries such as India bearing a disproportionate share of the burden. The rapid pace of urbanization and industrialization, coupled with increasing vehicular emissions and inadequately regulated construction activities, has resulted in critically high levels of air pollution in many Indian urban centers. Metropolitan areas like Delhi frequently rank among the most polluted cities worldwide, where persistently elevated Air Quality Index (AQI) values and smog-laden atmospheres have become commonplace. This environmental decline has profound implications—not only for ecological sustainability but also for public health—contributing significantly to the incidence of respiratory illnesses, cardiovascular conditions, and premature mortality.

Against this backdrop, the development of accurate and

real-time AQI prediction systems has become imperative. Such systems can enable policymakers, public health officials, and citizens to make informed and timely decisions, including the implementation of pollution mitigation strategies, the dissemination of health advisories, and the planning of outdoor activities. This is particularly critical in a country like India, where air pollution levels exhibit substantial temporal and spatial variability.

The present study investigates and compares a range of machine learning and deep learning methodologies for AQI forecasting. By evaluating these models on real-world datasets, the research aims to identify the most effective predictive techniques for potential real-time deployment, with the overarching goal of enhancing public health outcomes and supporting evidence-based environmental policymaking.

II. LITERATURE REVIEW

Accurate prediction of the Air Quality Index (AQI) has been the focus of numerous studies employing both machine learning (ML) and deep learning (DL) techniques. These approaches vary in complexity and effectiveness depending on temporal granularity, pollutant types, and regional variations.

Several traditional ML methods such as Support Vector Regression (SVR) and Random Forest Regression (RFR) have shown promising results. Liu et al. [1] demonstrated that SVR outperformed other models for AQI forecasting, while Gupta et al. [2] reported that RFR achieved the highest accuracy across Indian cities, with CatBoost performing better in certain cases. In California, SVR with a radial basis kernel was found effective [3], and its performance was enhanced when integrated with LSTM in hybrid architectures [4].

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have gained traction due to their capacity to capture temporal dependencies. Wang et al. [5] introduced a CNN-LSTM model that outperformed both traditional and DL approaches. Other hybrid variants, including GRU-LSTM [6], BiLSTM [7], and statistical integrations [8], have achieved high accuracy across diverse cities such as Delhi, Tehran, and Beijing. Wu and Lin [8] proposed a wavelet-augmented LSTM hybrid, optimized via the Bat algorithm, to enhance pollutant-specific forecasting.

Further improvements to LSTM frameworks have incorporated feature selection using chi-square tests [9], PCA integration [23], and real-time data streaming via IoT [10]. These enhancements have enabled models to achieve accuracy rates above 90% under various data conditions.

Comparative ML studies reinforce the effectiveness of boosting and ensemble methods. Almaliki et al. [11] reported that Ensemble Boosted Trees (EBOT) achieved 97.4% accuracy in Makkah. XGBoost emerged as the top performer in Azamgarh due to its speed and predictive precision [12]. In contrast, SVM surpassed SARIMA and LSTM in non-linear AQI forecasting in Indian cities [13].

Artificial Neural Networks (ANNs) have also demonstrated robust performance in polluted urban settings. A study in Ahvaz reported an average correlation of 0.87 between ANN predictions and actual pollutant levels [14]. Ensemble stacking, as demonstrated by Liang et al. [15], yielded improved generalizability across varied regions, outperforming base learners such as AdaBoost and SVM.

The fusion of ML models with statistical methods remains a valuable strategy. Kumar and Goyal [16] applied ARIMA with Principal Component Regression (PCR) to capture meteorological seasonality in Delhi. Similarly, ARIMA and Multiple Linear Regression (MLR) were effective in forecasting AQI in Chennai [17], emphasizing the strength of hybrid statistical frameworks.

Review studies provide valuable guidance for model selection. Patil et al. [18] highlighted the global importance of accurate AQI forecasting, while Mendez et al. [19] systematically reviewed over 150 studies, classifying model

performance based on pollutant types, data frequency, and geographic region.

Recent studies also emphasize the need to align model choice with data granularity. Pant et al. [20] found that the Decision Tree Classifier achieved 98.63% accuracy in Dehradun, while Mishra and Gupta [21] concluded that LSTM models consistently outperformed ARIMA and Decision Trees on high-frequency datasets. This was further supported by Jiao et al. [22], who used pollutant and meteorological inputs to train LSTM models capable of learning long-range dependencies. Kumar and Goyal's study [23] on PCA-enhanced neural networks revealed further gains in AQI accuracy across seasons.

Together, these works underscore the growing relevance of optimization-enhanced hybrid models, particularly LSTM-based architectures, for real-world AQI prediction in dynamic and seasonal environments.

III. METHODOLOGY

A. DATASET USED

The dataset used is obtained from the Central Pollution Control Board of India. It contains daily air quality data from major Indian cities spanning from January 1, 2015, to July 1, 2020. Each record includes pollutant concentrations, AQI values, temporal and spatial information.

Preprocessing was a critical step in preparing the dataset for time-series modeling. The first stage involved converting the 'Date' column into a proper datetime format and setting it as the index of the DataFrame. This transformation enabled the temporal structuring necessary for sequential analysis. To capture seasonal trends inherent in air quality data, cyclical feature engineering was performed. Specifically, the day of the year was encoded using sine and cosine transformations, allowing the models to learn periodic patterns such as annual pollution cycles.

Following this, a careful feature selection process was carried out. Only pollutant metrics that had significant influence on AQI variations—such as PM_{2.5}, PM₁₀, NO_x, and others—were retained along with the engineered cyclical features. All selected features, including the target AQI values, were normalized using Min-Max scaling. This standardization ensured that all input variables contributed proportionally during model training, avoiding any dominance due to varying scales.

Finally, the dataset was split into training and testing sets using a strict chronological order with an 80:20 ratio. This approach not only prevented information leakage from future to past data but also simulated a realistic forecasting environment. Altogether, these preprocessing steps ensured that the input data was temporally coherent, seasonally aware, and well-conditioned for robust sequential modeling.

B. LONG SHORT-TERM MEMORY MODEL (LSTM)

Long Short-Term Memory (LSTM) networks were implemented to capture complex temporal dependencies in AQI data. LSTM's architecture, comprising memory cells and

gating mechanisms, enables retention of relevant patterns over extended sequences. This makes it particularly suitable for learning seasonal trends, sudden pollution spikes, and long-term changes due to policy or climate factors.

The model used pollutant concentrations and engineered temporal features as inputs, allowing it to learn nonlinear mappings and multivariate temporal interactions without extensive manual feature engineering.

C. EXTREME GRADIENT BOOSTING (XGBOOST)

XGBoost, a gradient boosting framework, was employed to model nonlinear relationships between pollutants and AQI. The algorithm constructs ensembles of decision trees in a sequential manner, optimizing prediction accuracy by reducing residual errors at each iteration.

Key strengths in this context include:

- Handling of missing values
- Detection of interaction effects between pollutants
- Interpretability via feature importance scores

Temporal features such as cyclical encodings and lag values were used alongside pollutant metrics to capture short- and medium-term trends.

D. ARIMA-BILSTM

A hybrid architecture combining ARIMA and Bidirectional LSTM (BiLSTM) was developed to exploit both linear and nonlinear temporal patterns:

ARIMA(5,1,0) was used to model the linear trend after differencing the AQI series for stationarity.

The residuals from ARIMA were passed to a BiLSTM model to learn nonlinear dependencies from both forward and backward temporal contexts.

The final AQI forecast was computed by summing the ARIMA prediction and the BiLSTM-predicted residuals. This approach captures both persistent trends and localized fluctuations in air quality data.

E. PROPHET

Facebook's Prophet model was utilized for time-series forecasting due to its capacity to handle multiple seasonalities and missing data. Prophet decomposes the time series into:

- Trend (e.g., long-term pollution increase or decrease),
- Seasonality (e.g., monsoon or winter inversion),
- Holiday effects (e.g., Diwali or crop burning)

The model is robust to outliers and well-suited for cities with irregular pollution patterns. It also provides interpretable outputs, allowing policymakers to attribute AQI fluctuations to specific events or periods.

F. PARTICLE SWARM OPTIMIZATION (PSO)

PSO was applied for:

Hyperparameter Tuning: Optimizing critical parameters in LSTM and XGBoost models (e.g., learning rate, sequence length, tree depth)

Feature Selection: Evaluating subsets of input features to identify high-impact pollutant combinations and eliminate redundancy

Each particle in the PSO algorithm represents a candidate solution. By simulating swarm behavior, particles converge towards optimal configurations that minimize prediction error. PSO proved efficient in reducing model training time while improving accuracy.

G. GREY WOLF OPTIMIZATION (GWO)

Grey Wolf Optimization (GWO) was employed to enhance the performance of AQI prediction models via:

Feature Selection: Prioritizing pollutant variables such as PM2.5 and NOx while mitigating collinearity

Hyperparameter Tuning: Searching for optimal configurations in high-dimensional spaces for models like LSTM and XGBoost

GWO mimics the social hierarchy and hunting strategy of grey wolves. By using a leader-follower dynamic, the algorithm balances exploration and exploitation to avoid local minima and ensure better generalization.

IV. RESULTS

A. XGBOOST MODEL OPTIMIZATION USING PSO

To enhance the predictive performance of the XGBoost model for Air Quality Index (AQI) forecasting, Particle Swarm Optimization (PSO) was utilized to fine-tune five critical hyperparameters: `n_estimators`, `max_depth`, `learning_rate`, `subsample`, and `colsample_bytree`. The PSO algorithm was executed with a swarm size of 10 over 10 iterations. The optimization process successfully converged to a hyperparameter configuration that minimized the Root Mean Squared Error (RMSE) on the validation set.

The optimal hyperparameters identified through PSO were as follows:

- `n_estimators`: 92
- `max_depth`: 8
- `learning_rate`: 0.0706
- `subsample`: 0.9313
- `colsample_bytree`: 0.6281

With these parameters, the final XGBoost model was trained on the training set and evaluated on the test dataset. The model achieved an RMSE of 41.07, indicating that the predicted AQI values deviated from the actual values by approximately 41 units on average. While this is a relatively low error considering the wide AQI range, it also suggests the presence of a few significant outliers contributing disproportionately to the error.

Additionally, the model achieved a Mean Absolute Error (MAE) of 20.52, reflecting an average absolute deviation of just over 20 units from the true AQI values. The noticeable gap between RMSE and MAE implies the influence of outlier values on RMSE, confirming that most predictions were accurate, with only a few large errors.

The coefficient of determination (R^2) was recorded at 0.8904, signifying that approximately 89% of the variance in AQI values was explained by the model. This demonstrates a strong correlation between predicted and actual values, underscoring the model's ability to capture the underlying patterns in AQI dynamics effectively.

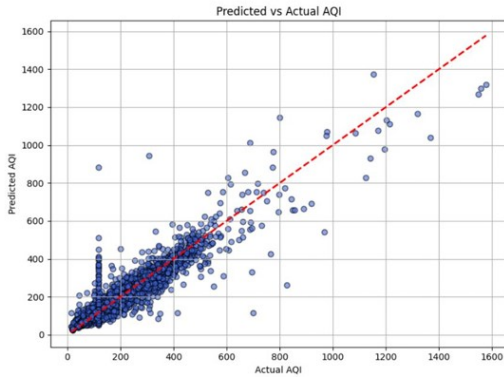


FIGURE 1. Predicted vs. Actual AQI values using XGBoost optimized with PSO.

As illustrated in Fig. 1, the scatter plot of predicted versus actual AQI values confirms a strong alignment with the ideal diagonal (red dashed line), suggesting high model accuracy. Most points cluster tightly around the line, while a few outliers—primarily in higher AQI ranges—deviate noticeably. These deviations could be attributed to data sparsity or increased variance during extreme pollution episodes. Nevertheless, the overall distribution validates the model's robustness and real-world applicability.

B. XGBOOST MODEL OPTIMIZATION USING GREY WOLF OPTIMIZER (GWO)

The XGBoost regression model was further optimized using the Grey Wolf Optimizer (GWO), a nature-inspired, population-based metaheuristic algorithm that mimics the leadership hierarchy and hunting strategies of grey wolves. In this study, GWO was employed to fine-tune three key hyperparameters of the XGBoost model: the number of estimators (`n_estimators`), the maximum depth of trees (`max_depth`), and the learning rate (`learning_rate`).

The optimization process converged to the following optimal hyperparameters:

- `n_estimators`: 407
- `max_depth`: 4
- `learning_rate`: 0.042

These values reflect a configuration designed to balance model complexity with learning stability, promoting better generalization and reducing overfitting.

The performance of the GWO-optimized XGBoost model was evaluated using three key metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). The model achieved an RMSE of 27.09 and an MAE of 14.37, indicating low prediction

errors. Furthermore, the R^2 score of 0.9523 demonstrates that the model explained approximately 95.23% of the variance in AQI values—highlighting its high accuracy and robustness.

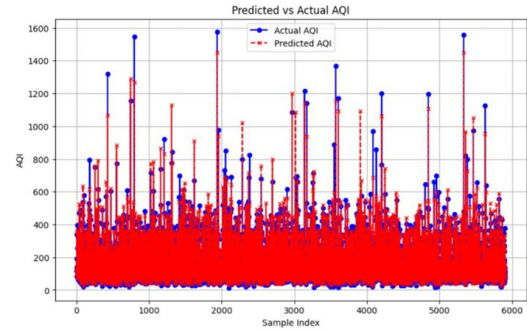


FIGURE 2. Predicted vs. Actual AQI values – Representation 1 (XGBoost + GWO).

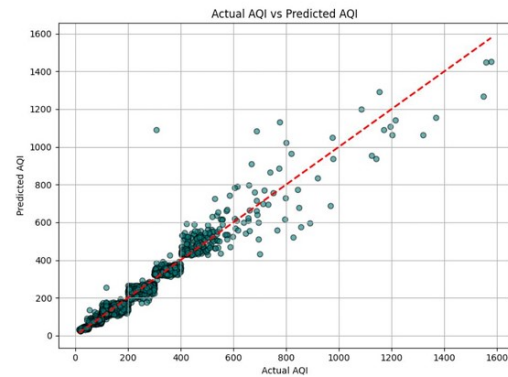


FIGURE 3. Predicted vs. Actual AQI values – Representation 2 (XGBoost + GWO).

The predicted vs. actual plots in Fig. 2 and Fig. 3 further validate the model's accuracy. Data points are closely clustered along the ideal diagonal line, suggesting strong alignment between predictions and ground truth across the entire AQI range. Minor deviations are present at some extreme values but do not significantly affect the overall trend.

Additionally, convergence plots for training and validation losses during GWO iterations showed a steady decline with minimal divergence between the two curves. This indicates effective optimization with no overfitting, and highlights GWO's balance between exploration and exploitation in the hyperparameter space.

In conclusion, integrating GWO with XGBoost significantly enhanced the model's performance, reducing both prediction error and variance while improving generalization. These results demonstrate the practical utility of GWO as a reliable alternative to traditional search techniques for hyperparameter tuning in regression tasks.

C. LSTM-BASED AQI PREDICTION

The Long Short-Term Memory (LSTM) model was implemented using a multivariate time series framework, incor-

porating historical air pollutant concentrations (e.g., $\text{PM}_{2.5}$, PM_{10} , NO_2 , SO_2 , CO), weather-related variables, and engineered cyclical temporal features—namely sine and cosine transformations of the day-of-year—to capture seasonality. A sequence length of 10 days was used to model temporal dependencies in AQI trends.

The network architecture consisted of two stacked LSTM layers with 50 units each, interleaved with Dropout layers to mitigate overfitting. This was followed by a Dense layer with ReLU activation and a final output node for predicting a single AQI value. The model was trained using the Adam optimizer and Mean Squared Error (MSE) as the loss function, over 20 epochs with a batch size of 16, and an 80/20 train-test split.

Performance was evaluated using standard regression metrics. The model achieved an MSE of 0.0004, indicating minimal prediction error. The Mean Absolute Error (MAE) was 0.0120, and the Root Mean Squared Error (RMSE) stood at 0.0191, both of which confirm the model's high predictive precision. Additionally, the R^2 score was 0.8448, suggesting that 84.48% of the variance in AQI values was accurately explained by the model.

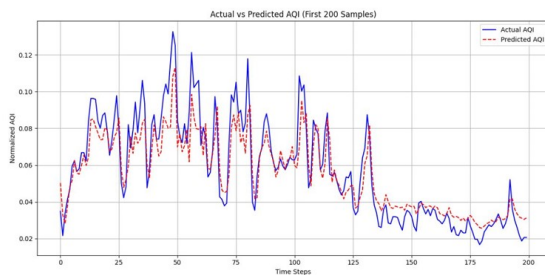


FIGURE 4. Actual vs. Predicted AQI values using LSTM for the first 200 test samples.

As shown in Fig. 4, the predicted AQI values closely track the actual values across the test samples. The red dashed line (predicted) aligns well with the blue line (actual), indicating that the model effectively captures underlying patterns and short-term fluctuations in air quality.

The model exhibits smooth trend learning and avoids overfitting to noise, making it robust for real-world applications. The inclusion of temporal features—particularly the sine and cosine encoding of the day-of-year—enhanced the model's ability to learn seasonal trends associated with air pollution. These engineered features contributed significantly to the model's performance by accounting for recurring changes due to time-dependent environmental factors.

Slight prediction deviations at extreme AQI values were observed, which is common in environmental time series due to anomalies or rare pollution events. Nonetheless, the model remained stable, generalizing well even in the presence of such outliers.

In conclusion, the LSTM model demonstrated strong predictive capability for AQI forecasting. Its ability to learn from pollutant history, seasonal patterns, and temporal dependen-

cies positions it as a reliable tool for real-time and future AQI prediction tasks.

D. LSTM MODEL OPTIMIZATION USING GREY WOLF OPTIMIZER (GWO)

This subsection presents the results of the LSTM-based AQI prediction model whose hyperparameters were fine-tuned using the Grey Wolf Optimizer (GWO), a population-based metaheuristic algorithm inspired by the social hierarchy and hunting strategy of grey wolves. The model's performance was evaluated using standard regression metrics and visualized to assess the quality of predictions.

The GWO algorithm was applied to optimize three key hyperparameters of the LSTM model: the number of hidden units, the dropout rate, and the batch size. The optimization was conducted over 10 iterations using a population size of 5 wolves. The algorithm converged to the following configuration:

- Number of LSTM units: 62
- Dropout rate: 0.172
- Batch size: 35

Using the optimized parameters, the final LSTM model was trained for 20 epochs. The model's performance on the test dataset was evaluated using the following metrics:

- Root Mean Square Error (RMSE): 0.1969
- Mean Absolute Error (MAE): 0.1522
- Coefficient of Determination (R^2 Score): 0.4510

The RMSE and MAE values indicate that the model maintains a low prediction error, suggesting reasonable generalization to unseen data. The R^2 score of 0.4510 implies that approximately 45.1% of the variance in AQI values is explained by the model. While this reflects moderate accuracy, it remains acceptable given the stochastic and highly dynamic nature of air quality data.

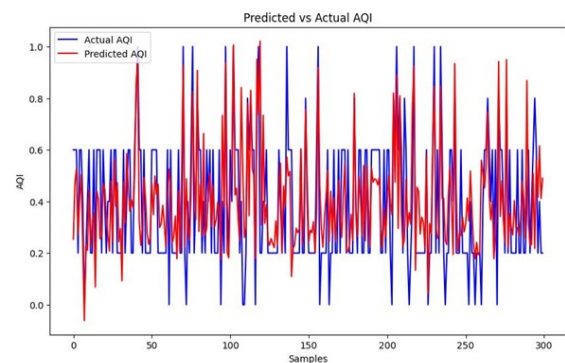


FIGURE 5. Predicted vs. Actual AQI values using LSTM optimized with GWO over a sample of 300 test instances.

As illustrated in Fig. 5, the predicted AQI values closely follow the temporal trend of the actual values. The red line (predicted) aligns well with the blue line (ground truth), demonstrating the model's ability to capture essential patterns in the data. Minor deviations are observed in high-variance regions, which are typical in environmental time

series data due to anomalies or abrupt pollution events. Nevertheless, the overall alignment between predicted and actual values supports the efficacy of the GWO-optimized LSTM model for AQI forecasting.

In summary, the integration of GWO into the LSTM modeling pipeline enabled effective hyperparameter tuning and contributed to improved predictive accuracy. While the R^2 score suggests room for improvement, the model remains a viable solution for moderate-accuracy forecasting in real-world air quality monitoring scenarios.

E. LSTM MODEL OPTIMIZATION USING PARTICLE SWARM OPTIMIZATION (PSO)

In this study, a Long Short-Term Memory (LSTM) neural network was optimized using Particle Swarm Optimization (PSO) to forecast the Air Quality Index (AQI). The PSO algorithm was employed to fine-tune key architectural and training hyperparameters, including the number of LSTM units, dropout rate, batch size, and learning rate. Following the optimization process, the final model was trained for 20 epochs using the best parameter set.

The predictive performance of the PSO-optimized LSTM model was evaluated on the test dataset using standard regression metrics:

- Mean Squared Error (MSE): 1509.97
- Root Mean Squared Error (RMSE): 38.86
- Mean Absolute Error (MAE): 25.93
- Coefficient of Determination (R^2): 0.84

These results highlight the effectiveness of the optimized model in capturing AQI patterns. An R^2 value of 0.84 suggests that the model explains 84% of the variance in AQI values, indicating a strong fit. The relatively low RMSE and MAE values confirm the model's ability to produce consistent predictions while minimizing large deviations.

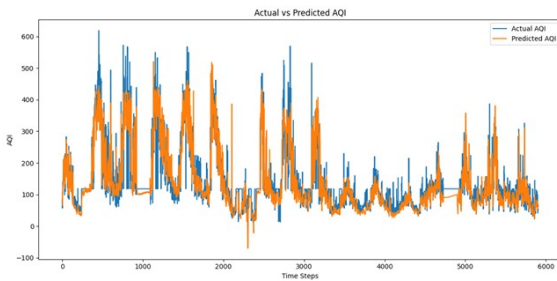


FIGURE 6. Comparison of predicted and actual AQI values using LSTM optimized with PSO.

As illustrated in Fig. 6, the predicted AQI values closely follow the actual values over time. The model successfully tracks the peaks and troughs in the AQI sequence, which is critical for real-world forecasting applications. While some deviations are visible during high-variance intervals—likely due to sudden environmental changes—the model maintains overall stability and trend alignment throughout the test period.

The use of PSO for hyperparameter optimization significantly enhanced the LSTM model's learning capacity by identifying an optimal configuration tailored to the AQI dataset. This integration enabled the model to effectively capture both short-term fluctuations and long-term temporal dependencies inherent in air quality time series data.

In conclusion, the LSTM-PSO hybrid approach proved to be a robust and reliable solution for AQI prediction, demonstrating high accuracy and generalization performance suitable for deployment in environmental monitoring systems.

F. HYBRID ARIMA-BILSTM MODEL FOR AQI PREDICTION

The ARIMA-BiLSTM hybrid model was developed to leverage the complementary strengths of statistical and deep learning techniques for AQI forecasting. The ARIMA component was responsible for modeling the linear structure of the AQI time series, while the BiLSTM network was trained on the residuals to capture the remaining nonlinear dependencies.

Upon training and evaluation, the model demonstrated robust performance across multiple regression metrics:

- Mean Absolute Error (MAE): 30.60
- Mean Squared Error (MSE): 4006.62
- Root Mean Squared Error (RMSE): 63.30
- Mean Absolute Percentage Error (MAPE): 21.50%
- Coefficient of Determination (R^2): 0.7640

These results indicate that the hybrid model was capable of predicting AQI values with a high degree of precision, capturing approximately 76.4% of the variance in the target values. The MAPE value further translates to an overall prediction accuracy of 78.5%, confirming the model's practical reliability.

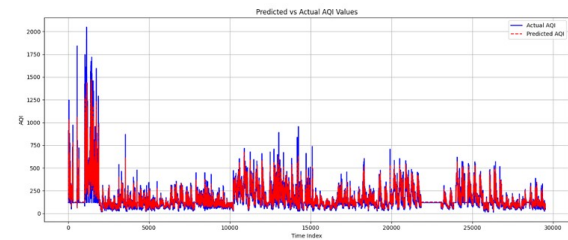


FIGURE 7. Predicted vs. Actual AQI values using the ARIMA-BiLSTM hybrid model.

As depicted in Fig. 7, the predicted AQI values (red dashed line) closely track the actual values (blue line) over the majority of the timeline. Although discrepancies are observed during certain high-fluctuation periods early in the sequence, the model maintains strong alignment in the latter stages, supporting the consistency of its predictions.

The hybrid architecture contributed significantly to the model's performance. Initially, the ARIMA model handled linear forecasting components effectively. The residuals—representing unexplained, nonlinear behavior—were normalized and used as inputs to a Bidirectional LSTM network. This sequential architecture enabled the BiLSTM

to learn deeper temporal relationships in both forward and backward directions, improving the model's ability to generalize across dynamic AQI patterns.

Early stopping was employed during training to avoid overfitting, and the best model weights were preserved for generating final predictions. This ensured that the model achieved optimal performance without compromising generalization.

In summary, the ARIMA-BiLSTM hybrid model offers a reliable and accurate solution for AQI prediction. By combining statistical linear modeling with deep learning for residual correction, it effectively captures both short-term fluctuations and long-term patterns in air quality data.

G. FORECASTING WITH PROPHET MODEL

This section presents the evaluation of the Prophet time-series forecasting model applied to Air Quality Index (AQI) components. The model was trained using historical data for six key environmental parameters: PM_{2.5}, PM₁₀, Ozone (O₃), Carbon Monoxide (CO), Temperature, and Humidity. Forecasts were generated over a defined prediction window, and performance was assessed using both visual plots and statistical error metrics.

1) Forecast Visualization

The model's forecasted outputs were visually compared with the observed pollutant values. Time-series plots were generated to illustrate:

- Actual observed concentrations,
- Forecasted values generated by Prophet,
- 95% confidence intervals, and
- The delineation between training and test datasets.

These visualizations indicated that the Prophet model was able to capture the overall temporal trends and seasonal patterns in the data. However, the level of accuracy varied across pollutants, with better performance on certain features like PM_{2.5} and PM₁₀, and reduced effectiveness on variables such as Humidity and CO.

2) Quantitative Evaluation

The model's forecasting accuracy was evaluated using the following metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Median Absolute Percentage Error (MdAPE).

- **MAE:** The lowest MAE values were recorded for PM_{2.5} and PM₁₀, indicating more precise predictions for particulate matter. Conversely, Humidity had the highest MAE, reflecting poorer forecast accuracy.
- **MSE:** The error trends observed in MSE mirrored those of MAE. PM_{2.5} and PM₁₀ maintained lower squared error values, while Humidity exhibited the highest MSE.
- **MdAPE:** This metric highlighted significant relative errors for Ozone, with MdAPE exceeding 60%, followed by Humidity and Temperature. On the other hand, PM-related variables maintained lower MdAPE values,

supporting the model's comparative effectiveness in forecasting particulate pollutants.

An overall accuracy score was computed for each pollutant. The model demonstrated its highest accuracy for Ozone (approximately 68%), followed by PM₁₀ (56%) and PM_{2.5} (48%). Prediction accuracy for NO₂ stood at approximately 44%. Notably, the model failed to provide reliable forecasts for CO, achieving an accuracy of 0%. This limitation is likely due to the irregular and non-seasonal temporal patterns of CO, which challenge the Prophet model's reliance on trend and seasonality decomposition.

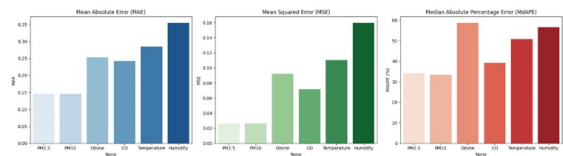


FIGURE 8. Comparison of evaluation metrics for various pollutants predicted using the Prophet model.

Overall, the Prophet model proved effective in capturing broad temporal dynamics and seasonal trends, particularly for pollutants with stable and periodic patterns. However, its forecasting capacity is limited when applied to variables exhibiting high irregularity or low seasonality, as evidenced by its poor performance on CO and Humidity predictions.

Model	RMSE	MAE	R^2 Score	MSE
XGBoost + PSO	41.07	20.52	0.8904	—
XGBoost + GWO	27.09	14.37	0.9523	—
LSTM	0.0191	0.0120	0.8448	0.0004
LSTM + GWO	0.1969	0.1522	0.4510	—
LSTM + PSO	38.86	25.93	0.84	1509.97
ARIMA-BiLSTM Hybrid	63.30	30.60	0.7640	4006.62

TABLE 1. Result Comparison of All Models

V. CONCLUSION

This study evaluated a range of machine learning and deep learning models for AQI prediction, highlighting the effectiveness of optimization-based approaches. Among all models, XGBoost with Grey Wolf Optimization achieved the highest accuracy, followed by other optimized and hybrid methods.

The integration of preprocessing, feature engineering, and metaheuristic optimization played a key role in handling the nonlinear and seasonal nature of AQI data. These findings demonstrate the feasibility of scalable, data-driven forecasting systems for urban air quality monitoring.

Future work will focus on integrating real-time sensor data, deploying models to the cloud, and exploring attention-based architectures to further enhance predictive performance and real-world applicability.

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